Active Methylene Compounds, Enols and Enolates

Active Methylene Compounds

Active methylene compounds contain a methylene group ($-CH_2-$) that is flanked by two electron-withdrawing groups (EWG), making the hydrogen atoms on the methylene carbon relatively acidic. These compounds can be deprotonated by bases to form stabilized carbanions.

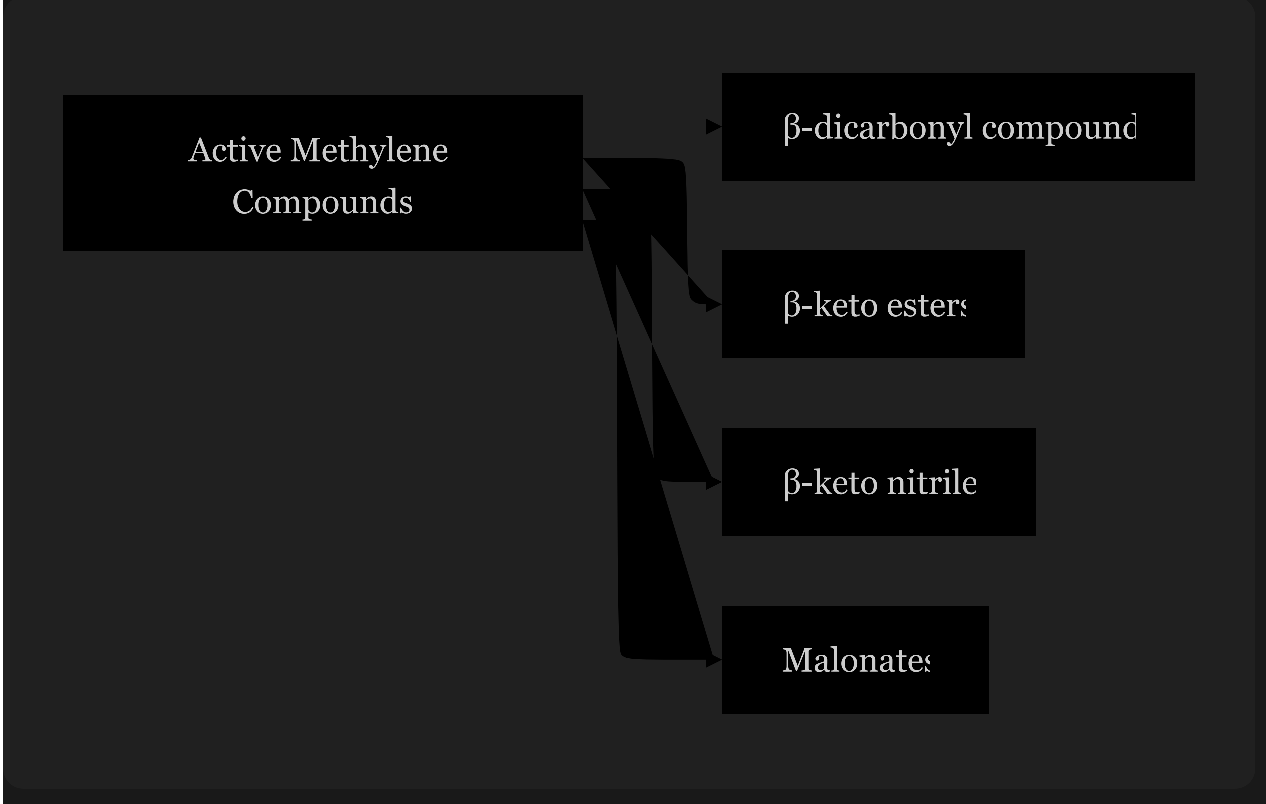
Common Examples:

β-dicarbonyl compounds (e.g., acetylacetone, diethyl malonate)

β-keto esters (e.g., ethyl acetoacetate)

β-keto nitriles

Malonates

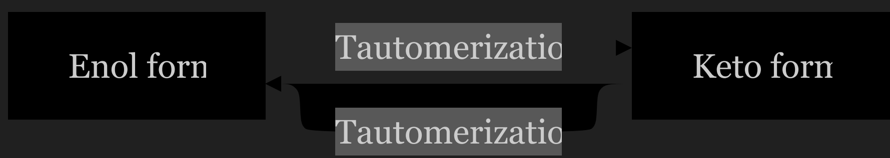


The acidity of these compounds is due to resonance stabilization of the resulting carbanion after deprotonation.

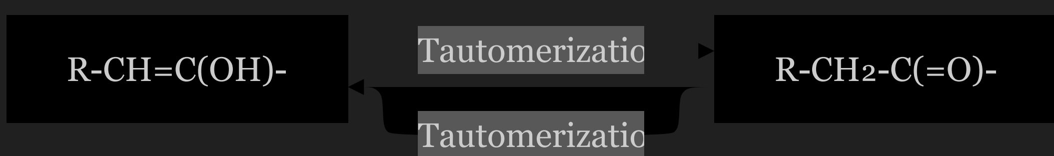
Enols and Enolates

Enols

Enols are alkenes that have a hydroxyl group (-OH) attached to one of the carbon atoms of the double bond. Carbonyl compounds can undergo tautomerization to form enols.



Keto-enol tautomerization:



Enolates

Enolates are the deprotonated form of enols or carbonyl compounds, forming a resonance-stabilized carbanion where the negative charge is delocalized between carbon and oxygen.

Kinetic and Thermodynamic Enolates

For unsymmetrical ketones, two different enolates can form depending on which proton is removed:

Kinetic Enolate:

Formed under kinetic control (low temperature, strong, bulky base like LDA)

Deprotonation occurs at the less substituted α-carbon (less sterically hindered)

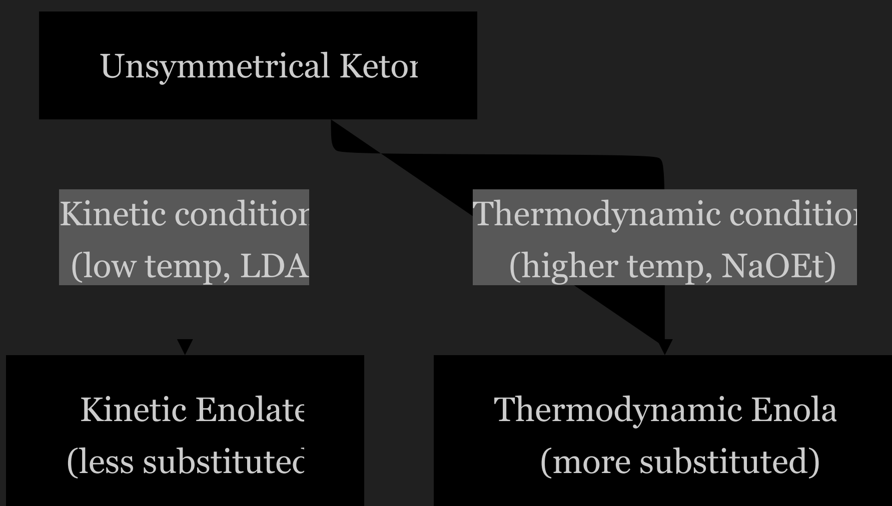
Represents the faster deprotonation pathway

Thermodynamic Enolate:

Formed under thermodynamic control (higher temperature, weaker base like NaOEt)

Deprotonation occurs at the more substituted α-carbon

More stable due to greater substitution at the double bond



Reactions of Active Methylene Compounds

Alkylation of Active Methylene Compounds

Alkylation involves the addition of an alkyl group to an active methylene compound through nucleophilic substitution. The reaction proceeds through the following steps:

Mechanism:

Deprotonation of the active methylene compound by a base to form a resonance-stabilized carbanion

Nucleophilic attack by the carbanion on the alkyl halide (SN2 mechanism)

Formation of the alkylated product

CH_3COCH_2COC → Base (NaOE → CH_3COCH^-COC → RX (e.g., CH → $CH_3COCH(R)COC$

Examples:

Diethyl malonate alkylation (Malonic ester synthesis):

$CH_2(COOEt$ → Unsupported markdown: l → $RCH(COOEt$ → Unsupported markdown: l → $RR'C(COOEt$ → Unsupported markdown: l → $RR'CHCOO$

Applications:

Synthesis of carboxylic acids via malonic ester synthesis

Preparation of functionalized ketones through acetoacetic ester synthesis

Synthesis of complex organic molecules in pharmaceutical and natural product chemistry

Acylation of Active Methylene Compounds

Acylation introduces an acyl group (R-C=O) to the active methylene carbon, typically using acyl halides or esters as acylating agents.

Mechanism:

Deprotonation of the active methylene compound by a base

Nucleophilic attack by the carbanion on the acylating agent

Formation of the acylated product

CH_3COCH_2COC → Base (NaH → CH_3COCH^-COC → RCOX (acyl halid → $CH_3COCH(COR)COC$

Examples:

Claisen condensation (self-condensation of esters):

2 CH_3COOI → NaOE → CH_3COCH_2COC → H_2O → CH_3COCH_2COC

Applications:

Synthesis of β-dicarbonyl compounds

Preparation of intermediates for heterocyclic compounds

Production of pharmaceutical intermediates

Halogenation of Active Methylene Compounds

Halogenation involves the introduction of halogen atoms (Cl, Br, I) to the active methylene position. This reaction can occur through various mechanisms depending on conditions.

Mechanisms:

1. Base-catalyzed halogenation:

Involves enolate formation followed by reaction with the halogen

Occurs under basic conditions

Typically gives monohalogenated products with controlled conditions

CH_3COCH → Base (OH^- → $CH_2=C(O^-)CH$ → X_2 (halogen → CH_2XCOCH

2. Acid-catalyzed halogenation:

Involves enol formation followed by electrophilic attack by the halogen

Occurs under acidic or neutral conditions

Can lead to polyhalogenation without careful control

CH_3COCH → H^+ → $CH_2=C(OH)CH$ → X_2 (halogen → CH_2XCOCH

3. Hell-Volhard-Zelinsky reaction:

A specific type of α-halogenation of carboxylic acids:

$$RCH_2COO \xrightarrow{X_2,\ PX} RCHXCOO$$

Detailed Example: Halogenation of Acetone

The halogenation of acetone with bromine in basic medium:

Step 1: Base (OH^-) removes a proton from acetone to form the enolate:

$$CH_3COCH_3 + OI \rightarrow CH_2{=}C(O^-)CH_3 + H$$

Step 2: The enolate attacks bromine, forming the α-brominated product:

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graph LR A["CH₂=C(O⁻)CH₃ + Br₂"] --> B["CH₂BrCOCH₃ + Br⁻"]
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$$CH_2{=}C(O^-)CH_3 + I \rightarrow CH_2BrCOCH_3 +]$$

Step 3: With excess base and bromine, further halogenation can occur:

$$CH_2BrCOCI \xrightarrow{OH^-} CHBr{=}C(O^-)CI \xrightarrow{Br_2} CHBr_2COCI \xrightarrow{OH^-,\ Br} CBr_3COCI$$

The Haloform Reaction

A special case of halogenation where methyl ketones react with halogens in base to form trihalomethanes and carboxylate salts:

CH_3CO $\xrightarrow{X_2,\ OH\ (excess}$ CX_3CO $\xrightarrow{OH^-}$ CHX_3 + RCO(

This reaction is particularly useful for:

- Identifying methyl ketones (iodoform test)
- Converting methyl ketones to carboxylic acids
- Synthesizing chloroform, bromoform, or iodoform

Factors Affecting Halogenation:

- pH of the reaction medium
- Type of halogen (reactivity: $I_2 < Br_2 < Cl_2$)
- Temperature and reaction time
- Presence of catalysts

Applications of Halogenated Active Methylene Compounds:

- Intermediates in organic synthesis
- Precursors for pharmaceutical compounds
- Building blocks for agrochemicals
- Reagents for further functional group transformations

Summary

Active methylene compounds are versatile substrates for a variety of synthetic transformations:

- Alkylation: introduces alkyl groups through nucleophilic substitution
- Acylation: introduces acyl groups through nucleophilic acyl substitution
- Halogenation: introduces halogen atoms, with mechanisms varying based on conditions

These reactions form the backbone of many synthetic strategies in organic chemistry, providing access to complex molecular structures from relatively simple starting materials.